

MOHAMED KHALED MAHMOUD

Full Stack .NET Developer

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Portfolio | linkedin.com/in/mohamed-khaled-16695a176 | github.com/kandeel11 | Military Service: Completed

PROFILE

Full Stack .NET Developer building scalable web applications with C#, ASP.NET Core, Angular 21, and SQL Server. Graduate of the Information Technology Institute (ITI) intensive Full Stack .NET program, holding an honors engineering degree ranked 2nd in class. Applies Clean Architecture, CQRS, and MediatR to build maintainable back-ends, shipped to Microsoft Azure through GitHub Actions CI/CD pipelines. Experienced in embedding AI into products — RAG pipelines and agentic workflows with Semantic Kernel — and in leading Agile teams through the full software lifecycle.

TECHNICAL SKILLS

- **Back-End:** C#, ASP.NET Core (Web API, MVC), Entity Framework Core, LINQ, RESTful APIs, JWT Authentication, ASP.NET Identity, Swagger/OpenAPI, SignalR, Dependency Injection, async/await
- **Architecture & Patterns:** Clean Architecture, CQRS, MediatR, Modular Monolith, SOLID Principles, Repository & Unit of Work, Design Patterns, OOP, Data Structures
- **Front-End:** Angular 21 (Standalone Components, Signals, RxJS, Lazy Loading, Reactive Forms), TypeScript, JavaScript (ES6+), HTML5, CSS3, Bootstrap
- **Databases:** Microsoft SQL Server, T-SQL, Query Optimization & Indexing, MySQL, Redis (Caching)
- **AI & Automation:** Semantic Kernel, RAG (Retrieval-Augmented Generation), Generative AI APIs, Prompt Engineering, Agentic Workflows, n8n, Python
- **Cloud, DevOps & Testing:** Microsoft Azure (App Service, Azure SQL Database), CI/CD with GitHub Actions, Docker, Git/GitHub, Postman, Agile/Scrum, NUnit, Moq
- **Languages:** Arabic (Native), English (Professional Working Proficiency)

EXPERIENCE & TRAINING

Information Technology Institute (ITI) – MCIT
Full Stack Web Development using .NET — Intensive Training Program (ITP)

Tanta, Egypt
Jan 2026 – Jul 2026

- Built RESTful back-end services with C# and ASP.NET Core, applying Clean Architecture, CQRS, and MediatR to keep business logic decoupled and testable.
- Led a 5-developer Agile team through 4-week sprints on the graduation project, managing a Git feature-branch workflow with pull-request reviews and containerizing services with Docker.
- Developed responsive front-ends with Angular 21 and TypeScript using standalone components, Signals, and reusable UI modules.
- Designed relational schemas in SQL Server with Entity Framework Core, tuning LINQ queries and indexes for high-performance data retrieval.

Egyptian Armed Forces
Mandatory Military Service — Completed

Egypt
Nov 2024 – Dec 2025

- Completed mandatory national service while continuing self-directed study in .NET and Angular.

Outlier
AI Model Trainer (Freelance)

Remote
May 2024 – Oct 2024

- Evaluated generative AI model outputs across 50+ review cycles, scoring accuracy and logical reasoning against structured rubrics.
- Produced training data for large-scale datasets through prompt engineering and precise data annotation.

Information Technology Institute (ITI) – MCIT
Backend Web Development using .NET — Summer Code Camp

Mansoura, Egypt
Jul 2023 – Aug 2023

- Built RESTful APIs with ASP.NET Core and C#, integrated end-to-end with front-end components.
- Implemented data access with Entity Framework Core, reducing redundant queries through projection and eager loading, and validated endpoints with Postman collections.

PROJECTS

AI-Powered Real Estate Management Platform | ASP.NET Core, Angular 21, Redis, Semantic Kernel, Azure OpenAI
[Live Demo](#) | [GitHub](#)

- Architected a property management system as a Modular Monolith with CQRS and MediatR, keeping modules loosely coupled and independently deployable.
- Implemented Redis caching with write-invalidation on high-traffic listing endpoints, serving repeated queries from cache instead of the database.
- Built an AI assistant with Semantic Kernel that answers property queries via RAG over listing documents, and automated approval workflows with n8n.
- Deployed to Microsoft Azure (App Service + Azure SQL Database) with a GitHub Actions CI/CD pipeline handling build and

deployment on every push to main.

Vortex — Multi-Vendor E-commerce Platform | *ASP.NET Core, Angular 21, SQL Server, Stripe/PayPal* [GitHub](#)

- Designed the core database schema and built the full RESTful API for the product catalog, cart, and checkout pipeline using ASP.NET Core.
- Developed role-based dashboards (Admin, Seller, Customer) in Angular 21 with reusable components and route guards.
- Integrated Stripe and PayPal for secure payments and managed the complete order lifecycle with automated stock deduction.

Ecobazar — Multi-Role Organic Grocery Platform | *JavaScript (ES6+), Bootstrap 5, HTML5, CSS3* [Live Demo](#) | [GitHub](#)

- Led a 4-developer team as Team Leader, coordinating task distribution and delivery across the full build.
- Built a three-role platform (Customer, Seller, Admin) with product browsing, cart and wishlist, order tracking, seller inventory management, and admin account controls.
- Modelled the domain with ES6 classes and handled persistence and cross-tab synchronisation through localStorage and sessionStorage, with a fully responsive Bootstrap 5 UI.

AI-Powered Floor Plan Generator (Graduation Project) | *Python, React, Three.js, GANs* [GitHub](#)

- Developed an automated architectural layout generator using Python and GANs to optimize space utilization.
- Exposed the model through RESTful APIs, processing user constraints and rendering 3D results in under 2 seconds.
- Automated the initial sketch phase with computer vision, removing manual drafting from the early design workflow.

EDUCATION

Higher Institute of Engineering and Technology

B.Sc. in Computer and Control Engineering

- Grade: Excellent with Honors — Ranked 2nd in the 2024 graduating class.

Kafr El-Sheikh, Egypt

Aug 2019 – Jul 2024